# Amar Shah

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## Research Interests

My background is in formal verification, automated reasoning, and their interplay. Specifically, I am interested in building better tools to reason about programs and applying them to verifying critical software. Recently, I have been thinking about applications to AI safety.

## Education

**Carnegie Mellon University** August 2024 - Present
PhD in Computer Science advised by Marijn Heule & Bryan Parno

**University of California, Berkeley** January 2020 - December 2023
BA Computer Science and Math; Minor in Logic GPA: 3.90
Honors Thesis: *Reasoning about Algebraic Datatypes* advised by Federico Mora & Sanjit A. Seshia

## Research Internships

Amazon Web Services (Applied Science Intern) May 2025 - January 2026, May 2026 - January 2027

- Designed and built Sundance, an SMT solver for program verification, from scratch in Rust
- Competitive with existing SMT solvers, while being small, modular, and proof-producing
- Paper under submission [1] and we are working on deploying this in production

Advanced Software Technologies Group (ETH Zürich) February 2024 - July 2024

- Built fuzzing and testing for database systems and SMT solvers

## Refereed Conference & Journal Publications

[1] **Amar Shah**, Benjamin F. Jones, Can Cebeci, Jason Z. S. Hu, Lee A. Barnett, Rèmi Delmas, Marijn Heule, Bryan Parno, and Bruno Dutertre. "Sundance: A Simple SMT solver for Program Verification". In: *Under Submission*. 2026.

[2] Zachary Kent* and **Amar Shah***. "Parallelizing Congruence Closure". In: *Formal Methods in Computer-Aided Design (FMCAD)*. 2026.

[3] **Amar Shah**, Twain Byrnes, Joseph Reeves, and Marijn J.H. Heule. "Learning Short Clauses via Conditional Autarkies." In: *Formal Methods in Computer-Aided Design (FMCAD)*. Oct. 2025.

[4] Yi Zhou, **Amar Shah**, Zhengyao Lin, Marijn J.H. Heule, and Bryan Parno. "Cazamariposas: Automated Instability Debugging in SMT-based Program Verification". In: *Proceedings of the Conference of Automated Deduction (CADE)*. July 2025.

[5] **Amar Shah**, Federico Mora, and Sanjit A. Seshia. "An Eager Satisfiability Modulo Theories Solver for Algebraic Datatypes". In: *Thirty-Eighth AAAI Conference on Artificial Intelligence*. Jan. 2024.

[6] Jiaxin Yuan, **Amar Shah**, Channing Bentz, and Maria Cameron. "Optimal control for sampling the transition path process and estimating rates". In: *Communications in Nonlinear Science and Numerical Simulation (CNSNS)* (2024).

## Refereed Workshop Publications

[1] **Amar Shah**, Marijn Heule, Bryan Parno, and Max Willsey. "Relational E-matching in an SMT solver". In: *EGRAPHS Workshop at PLDI*. 2026.

[2] **Amar Shah**, Yi Zhou, Marijn J.H. Heule, and Bryan Parno. "Instability Track for SMT-COMP". In: *Satisfiability Modulo Theories (SMT)* (2025).

## Open Source Software

**Sundance** Primary developer

- An SMT Solver for Program Verification

**CaDiCaL** Primary developer

- A fork of the SAT solver CaDiCaL with conditional autarkies

**Algaroba** Primary developer

- An Eager SMT Solver for the theory of Algebraic Datatypes
- **Winner of QF_Datatypes division at SMTCOMP 2024**

## Teaching Experience

**Guest Lectures**

15-816: Advanced Topics in Logic: Automated Reasoning and Satisfiability
- Satisfiability Modulo Theories

MathIly-Er High School Camp
- Quantum Logic Gates

**Teaching Assistant**

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| 15-816: Advanced Topics in Logic: Automated Reasoning and Satisfiability | Fall 2025 |
| Math 1A: Calculus | Fall 2022 |
| Berkeley Math Circle Elementary | Fall 2021, Spring 2022 |
| CS 70: Discrete Mathematics & Probability Theory | Summer 2021 |

**Course Reader**

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| C 191: Quantum Information Science & Technology | Spring 2023 |
| CS 170: Efficient Algorithms & Intractable Problems | Spring 2023 |
| CS 70: Discrete Mathematics & Probability Theory | Fall 2021, Spring 2022 |

## Awards

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| Gold Medal QF_Datatypes division at SMTCOMP | 2024 |
| Second Place Winner ACM Grand Finals, Undergraduate Student Research Competition | 2024 |
| First Place Winner, PLDI Undergraduate Student Research Competition | 2023 |
| EECS Evergreen Research Award (UC Berkeley) | 2023 |
| Summer Undergraduate Research Fellow (UC Berkeley) | 2023 |
| Math Outstanding Graduate Student Instructor (UC Berkeley) | 2023 |
| Gulf Coast Undergraduate Research Symposium Outstanding Presentation | 2022 |
| Eagle Scout | 2019 |

## Academic Service

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| TACAS Artifact Evaluation Committee | 2027 |
| CMU Cylab Community Committee Member | Fall 2025 |
| PhD Student Introductory Course Organzier | Fall 2025 |
| PhD Student Orientation Organizer | Spring 2024 |
| Graduate Application Support Program Mentor | Fall 2024, Fall 2025 |
| Berkeley Undergraduate Mathementoring Program Mentor | Fall 2021, Fall 2023 |

## Miscellaneous

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| Programming Languages: | Python; C++; Rust; OCaml; Lean; LaTeX; Assembly (RISC-V) |
| Tools: | SMT solvers (Z3 & cvc5); Interactive Theorem Provers (Lean, Rocq, Agda) Deductive Verifiers (Verus, Dafny) |
| Languages: | English (native), Hindi (fluent), Spanish (basic), German (basic) |